FLUCLOXACILLIN

Cholestatic jaundice

Empty stomach (PO)

CHEMICAL NAME (2S,5R,6R)-6-[[3-(2-chloro-6-fluorophenyl)-5-methyl-1,2-oxazole-4-carbonyl]amino]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid (sodium)

CLASS Penicillinase-resistant (anti-staphylococcal) penicillin

TRADE NAMES Floxapen · Flucil · Staphylex

THERAPEUTIC USES / INDICATIONS

- Staphylococcal (MSSA) skin and soft-tissue infections: cellulitis, impetigo, wound infections, infected eczema.
- Osteomyelitis and septic arthritis.
- Staphylococcal endocarditis, pneumonia (IV).

ADMINISTRATION

- **PO:** 1 h before food or on an empty stomach.
- **IV:** slow injection over 3–4 min or infusion.
- **IM** injection.

DOSAGES

- **Adult PO:** 500 mg–1 g every 6 h.
- **Adult IV:** 1–2 g every 6 h (endocarditis up to 2 g every 4 h).
- **Children PO:** 2–10 yr 125 mg four times daily; <2 yr 62.5 mg four times daily. **IV/IM (<14 yr):** 25–50 mg/kg/day in 3–4 doses.
- Reduce in severe renal impairment.

PHARMACOKINETICS

- **Peak:** PO 1 h. Better oral absorption than cloxacillin.
- Highly protein-bound; partly metabolised in liver.
- Excreted in urine. **t½:** about 1 h.

PHARMACODYNAMICS

- β-lactamase-stable penicillin → inhibits cell-wall synthesis of **penicillinase-producing staphylococci**.
- Active against MSSA and streptococci; not MRSA.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea, dyspepsia.
- Rash.
- Phlebitis.

ADVERSE EFFECTS (SERIOUS)

- Cholestatic jaundice and hepatitis (may occur up to 2 months after stopping; more in >50 yr and courses >14 days).
- Anaphylaxis.
- Neutropenia, interstitial nephritis.
- *C. difficile* colitis.

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- History of flucloxacillin-associated jaundice.
- **Caution:** hepatic impairment, elderly.

KEY INTERACTIONS

- **Paracetamol (long-term, high dose):** high anion gap metabolic acidosis (rare).
- **Methotrexate:** ↑ toxicity.
- **Warfarin:** ↓ INR possible.

NURSING MANAGEMENT

- Check allergy history.
- Give oral doses on empty stomach.
- Monitor LFTs in prolonged courses or elderly.
- Watch for jaundice, dark urine, pale stools.
- Monitor wound/infection response and cultures.

PATIENT EDUCATION

- Take on an empty stomach, 4 times a day.
- Report yellow skin or eyes, dark urine — even weeks after finishing.
- Complete the full course.

KEY POINTS

- Anti-staphylococcal penicillin of choice (MSSA).
- Cholestatic jaundice risk, may be delayed.
- Empty stomach orally.
- Not effective against MRSA.

SOURCES CHECKED **[119]** Flucloxacillin 1000 mg powder for solution for injection or infusion (UK SmPC) **[120]** Flucloxacillin 500 mg capsules, hard (UK SmPC) **[121]** Flucloxacillin 125mg/5ml Powder for Oral Solution (UK SmPC)

Full references p. 432–457